



Answer confidently in the Interviews How to Evaluate the Accuracy of LLM Outputs

A Comprehensive Guide with Techniques, Examples, and Code



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Why Evaluate LLM Outputs?



- **To ensure reliability:** Check if the model consistently provides accurate answers.
- **To build trust:** Users need confidence in the outputs.
- **To identify areas of improvement:** Highlight where the model can perform better.
- **Relevance:** Outputs must align with the context or task.

Techniques Overview

- 1. Exact Match (EM)**
- 2. Precision, Recall, and F1 Score**
- 3. BLEU Score**
- 4. ROUGE Score**
- 5. Perplexity**
- 6. Human Evaluation**
- 7. Use Cases Testing**
- 8. User Feedback**
- 9. Calibration Testing**
- 10. Robustness Testing**
- 11. Adversarial Testing**

Exact Match (EM)

Definition: Compares the LLM output word-for-word with the expected correct answer.

How to use:

- Count how many answers are exact matches.
- Divide by the total number of questions.

$$\text{EM} = \frac{\text{Number of Exact Matches}}{\text{Total Questions}} \times 100$$

Example:

- Correct Answer: "Paris"
- LLM Output: "Paris" → Match
- LLM Output: "France" → No Match

Precision, Recall, and F1 Score

Precision: Measures how many provided answers are correct

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

Recall: Measures how many correct answers were found.

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

F1 Score: Balances precision and recall

$$\text{F1 Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

BLEU Score

$$\text{BLEU} = BP \times \exp \left(\sum_{n=1}^N w_n \log p_n \right)$$

Where:

- **BP (Brevity Penalty):** A penalty to prevent shorter outputs from getting higher scores.

$$BP = \begin{cases} 1 & \text{if } c > r \\ e^{(1-r/c)} & \text{if } c \leq r \end{cases}$$

- c : Length of candidate text.
- r : Length of reference text.
- p_n : Precision for n-grams (e.g., unigrams, bigrams).
- w_n : Weight assigned to each n-gram (default is equal weighting).

Definition: Measures how close the LLM's output is to a reference text.

Example:

- Reference: "The cat sat on the mat."
- LLM Output: "The cat is on the mat."
- BLEU Score: High (similar word structure).

ROUGE Score

ROUGE-N Formula

$$\text{ROUGE-N} = \frac{\sum_{n\text{-gram} \in \text{reference}} \text{Count}_{\text{match}}}{\sum_{n\text{-gram} \in \text{reference}} \text{Count}_{\text{reference}}}$$

Where:

- ***n*-gram**: Sequences of *n* words.
- **Count_{match}**: Number of *n*-grams in both the reference and candidate.
- **Count_{reference}**: Total *n*-grams in the reference.

ROUGE-L Formula

$$\text{ROUGE-L} = \frac{LCS(\text{candidate}, \text{reference})}{\text{Length of reference}}$$

Where:

- **LCS**: Longest Common Subsequence.

Definition: Measures the overlap of important words between the LLM's output and the reference.

Common Usage: Text summarization tasks.

Perplexity

```
!pip install rouge
```

```
Collecting rouge
```

```
  Downloading rouge-1.0.1-py3-none-any.whl.metadata (4.1 kB)
```

```
Requirement already satisfied: six in /usr/local/lib/python3.10/dist-packages (from rouge) (1.17.0)
```

```
Downloading rouge-1.0.1-py3-none-any.whl (13 kB)
```

```
Installing collected packages: rouge
```

```
Successfully installed rouge-1.0.1
```

```
from nltk.translate.bleu_score import sentence_bleu
```

```
from rouge import Rouge
```

```
# BLEU Score Example
```

```
reference = ["The cat sat on the mat"]
```

```
candidate = "The cat is on the mat"
```

```
bleu_score = sentence_bleu([reference], candidate.split())
```

```
print(f"BLEU Score: {bleu_score}")
```

```
# ROUGE Score Example
```

```
rouge = Rouge()
```

```
scores = rouge.get_scores(candidate, reference[0])
```

```
print(f"ROUGE Scores: {scores}")
```

```
BLEU Score: 0
```

```
ROUGE Scores: [{'rouge-1': {'r': 0.8333333333333334, 'p': 0.8333333333333334, 'f': 0.8333333283333335},
```

Definition: Indicates how "confident" the LLM is when generating an answer.

$$PPL = 2^{-\frac{1}{N} \sum_{i=1}^N \log_2 P(w_i)}$$

Lower Perplexity: Better predictions.

Disclaimer:

The code provided is solely for demonstration purposes to illustrate syntax and concepts. For accurate and up-to-date information, please refer to the official Python documentation.

Human Evaluation



Definition: Experts review the LLM's output for quality and sense.

Example:

- Question: "Why is the sky blue?"
- LLM Answer: "Because molecules scatter blue light more than other colors."
- Human Feedback: Correct and understandable.

Use Case Testing



Test the LLM on real-world applications:

- **Example 1:** Factual Answers: "Who is the President of the USA?"
- **Example 2:** Creative Writing: "Write a poem about the ocean."

User Feedback



Definition: Collect feedback from end users about LLM performance.

Why Important? Provides insights into usability and relevance.

Example: Feedback from students using LLM for homework.

Calibration Testing

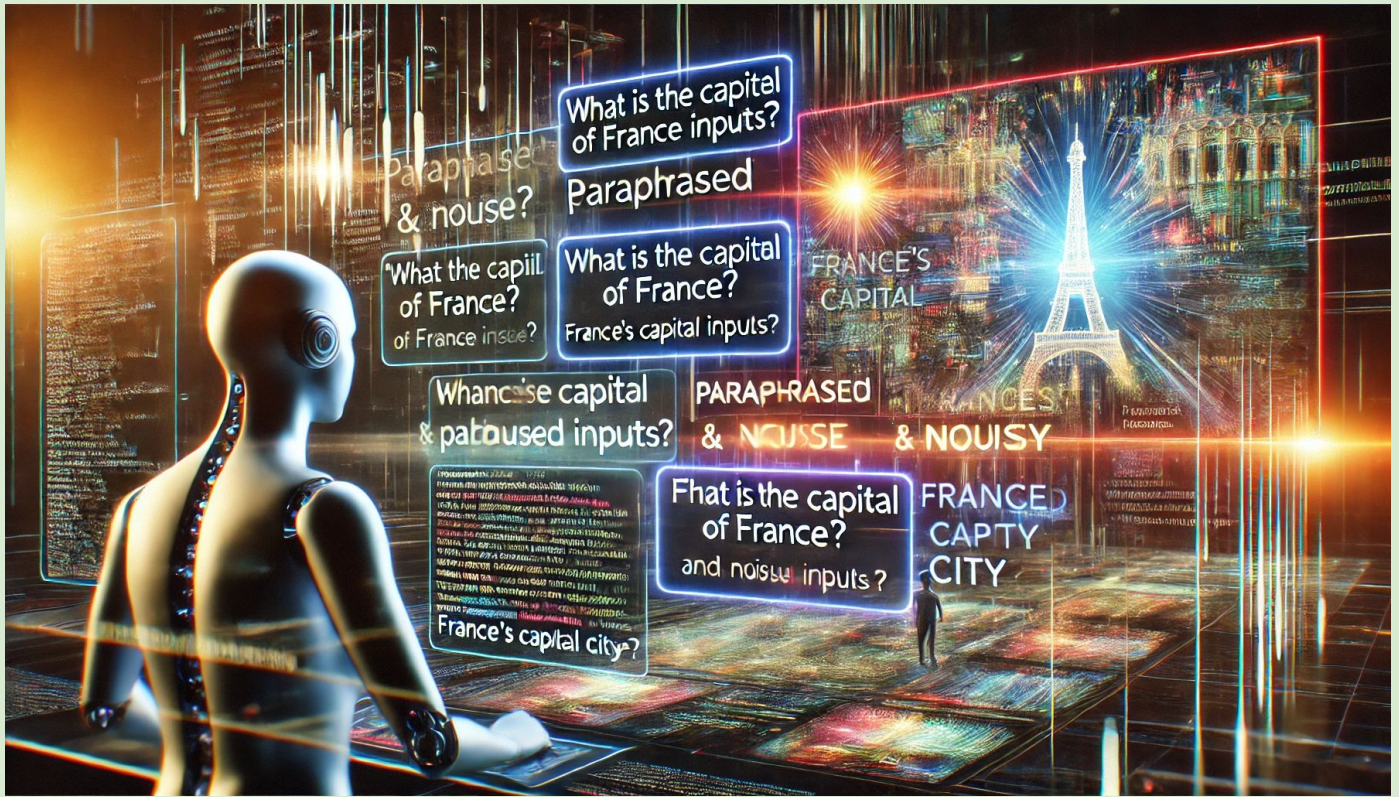


Definition: Tests if the model's confidence in its predictions matches its accuracy.

Example:

- LLM says: "I'm 90% sure the answer is Paris."
- Check if such answers are correct 90% of the time.

Robustness Testing



Definition: Check if the LLM works well even with:

- Paraphrased inputs.
- Noisy data.

Example:

- Original Question: "What is the capital of France?"
- Paraphrased: "France's capital city?"

Adversarial Testing



Definition: Expose the model to tricky questions.

Example:

- Question: "What is heavier: 1 kg of feathers or 1 kg of steel?"
- Correct Answer: "They weigh the same."

Combining Scores

```
bleu = 0.85
human_feedback = 0.9
f1_score = 0.8

final_score = 0.4 * bleu + 0.3 * human_feedback + 0.3 * f1_score
print(f"Final Evaluation Score: {final_score}")
```

Final Evaluation Score: 0.8500000000000001

Use weighted scoring for final evaluation.

- BLEU: 40%
- Human Feedback: 30%
- F1 Score: 30%

Formula:

$$\text{Final Score} = 0.4 \times BLEU + 0.3 \times \text{Human Feedback} + 0.3 \times \text{F1 Score}$$

Real-Life Analogy



Evaluating a Recipe:

1. **Exact Match:** Did the cook follow the recipe?
2. **BLEU/ROUGE Scores:** Are the ingredients similar?
3. **Human Feedback:** Does it taste good?
4. **Robustness:** Can you still cook it if instructions are jumbled?

Conclusion



- No single metric is perfect—combine methods.
- Use objective metrics for numbers and subjective metrics for meaning.
- Test in diverse, real-world scenarios.
- Continuously improve based on feedback.

Unlock the Power of Your Data

From patterns to possibilities, let's turn your data into impact.

Ready to elevate your outcomes? Let's shape the future together with actionable insights.



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